

Machine Learning: Advanced Techniques



I Generalization

I.1 Basics & Necessity of Assumptions

Werner Zellinger – Institute for Machine Learning

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Recap



AI Curriculum

- ML: Basic Techniques
Understand basic ML techniques
 - Classification, regression, uncertainty, regularization, ...
- ML: Supervised/Unsupervised Techniques
Use and choose ML techniques
 - supervised: NNs, SVMs, ...
 - unsupervised: clustering, projection, data reduction, ...
- **ML: Advanced Techniques**
Understand, design and modify ML techniques
 - generalization, inference, representation, dynamics

Content

- **Part I Generalization** When does learning work?
V. Vapnik, *The Nature of Statistical Learning Theory*, 2000.
- **Part II Inference** What is optimal under uncertainty?
H. Bishop, *Pattern Recognition and Machine Learning*, 2006.
- **Part III Representation** What functions/distributions can be expressed?
J. Tomczak, *Deep Generative Modeling*, 2022.
- **Part IV Learning dynamics** How do algorithms reach solutions?
S. Sra, S. Nowozin, and St. Wright, *Optimization for Machine Learning*, 2012.

Simple Learning Recipe

```
1  # data ~ p
2  X_train, y_train = load_data()
3  # part III representation: choose model class
4  model = Model(...)
5  # part II inference: define what should be estimated
6  objective = lambda theta: mean(loss(model(theta, X_train), y_train))
7  # part IV learning dynamics: compute estimator
8  theta_hat = Optimizer(...).minimize(objective)
9  # part I generalization: evaluate unseen performance
10 test_error = evaluate(model(theta_hat))
```

Part I: Generalization

I.1 Basics & necessity of assumptions

Risk, generalization, concentration, no free lunch

I.2 Vapnik-Chervonenkis theory

Consistency, uniform convergence, generalization bounds

I.3 Modern aspects

Double descent, distribution shifts

Recap of Basics



Linear Algebra Notation

- **Scalar**: $\alpha \in \mathbb{R}$
- **Vector**: $\mathbf{x} = (x_1, \dots, x_d)^\top \in \mathbb{R}^d$
- **Matrix**: $\mathbf{X} \in \mathbb{R}^{l \times d}$ with rows $\mathbf{x}_1^\top, \dots, \mathbf{x}_l^\top$

$$\mathbf{X} = \begin{pmatrix} \mathbf{x}_1^\top \\ \vdots \\ \mathbf{x}_l^\top \end{pmatrix}$$

- **Inner product**: $\langle \mathbf{x}, \mathbf{z} \rangle = \mathbf{x}^\top \mathbf{z} = \sum_{j=1}^d x_j z_j$
- **Euclidean norm**: $\|\mathbf{x}\|_2 = \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle}$

Probability Space and Expectation

- **Probability space:** $(\Omega, \mathcal{A}, \mathbb{P})$
- **Real random variable/vector:** $\mathbf{x} : \Omega \rightarrow \mathbb{R}^d$
- **Joint distribution:** $(\mathbf{x}, \mathbf{y}) \sim p(\mathbf{x}, \mathbf{y})$, with factorization $p(\mathbf{x}, \mathbf{y}) = p(\mathbf{x}) p(\mathbf{y}|\mathbf{x})$
- **Expectation** of a random variable $g(\mathbf{x}, \mathbf{y})$: $\mathbb{E}_{(\mathbf{x}, \mathbf{y}) \sim p}[g(\mathbf{x}, \mathbf{y})]$
- Special case (indicator): $\mathbb{E}[\mathbf{1}\{A\}] = \mathbb{P}(A)$
- For discrete distributions: $\mathbb{E}[g(\mathbf{x}, \mathbf{y})] = \sum_{\mathbf{x}, \mathbf{y}} g(\mathbf{x}, \mathbf{y}) p(\mathbf{x}, \mathbf{y})$

IID Sampling

- **Dataset:** $\mathbf{Z} = ((\mathbf{x}_1, y_1), \dots, (\mathbf{x}_l, y_l))$
- **IID assumption:** $(\mathbf{x}_i, y_i)_{i=1}^l$ are i.i.d. samples (realizations of independent copies of $(\mathbf{x}, y)$) from $p(\mathbf{x}, y)$
- Equivalently: $\mathbf{Z} \sim p^l$
- **Marginals:** $\mathbf{x}_i \sim p(\mathbf{x})$ and $y_i \sim p(y|\mathbf{x}_i)$
- Key property: $\mathbb{P}((\mathbf{x}_1, y_1), \dots, (\mathbf{x}_l, y_l)) = \prod_{i=1}^l p(\mathbf{x}_i, y_i)$

Dataset, Loss and Risk

- Binary classifier $f : X \rightarrow \{-1, 1\}$ and zero-one loss

$$L(y, y') = \mathbf{1}\{y \neq y'\} = \begin{cases} 0 & \text{if } y = y' \\ 1 & \text{if } y \neq y' \end{cases}$$

- Dataset $\mathbf{Z} = (\mathbf{x}_i, y_i)_{i=1}^l$ are i.i.d. samples from $p(\mathbf{x}, y)$
- Expected risk

$$R(f) = \mathbb{E}_{(\mathbf{x}, y) \sim p}[L(y, f(\mathbf{x}))]$$

- Empirical risk

$$R_{\text{emp}}(f) = \frac{1}{l} \sum_{i=1}^l L(y_i, f(\mathbf{x}_i))$$

Notation

- **Bayes predictor** $f^* \in \arg \min_f R(f)$ has minimal expected risk
 - For binary classification:

$$f^*(\mathbf{x}) = \begin{cases} +1 & p(y = +1|\mathbf{x}) > p(y = -1|\mathbf{x}) \\ -1 & p(y = -1|\mathbf{x}) > p(y = +1|\mathbf{x}) \\ \text{either} & \text{otherwise} \end{cases}$$

- Goal: $R(\hat{f}) \approx R(f^*)$
- **Hypothesis class** $\mathcal{F}$
- **Best-in-class predictor**

$$f_{\mathcal{F}} \in \arg \min_{f \in \mathcal{F}} R(f)$$

- **Learned hypothesis** $\hat{f} = A(\mathbf{Z})$ with $A : \mathcal{Z}^l \rightarrow \mathcal{F}$

Error Decomposition

- Goal: compare learned predictor $\hat{f}$ with Bayes predictor f^*
- Excess risk decomposition:

$$R(\hat{f}) - R(f^*) = \underbrace{R(\hat{f}) - R(f_{\mathcal{F}})}_{\text{estimation error}} + \underbrace{R(f_{\mathcal{F}}) - R(f^*)}_{\text{approximation error}}$$

- Estimation error
 - due to finite sample size
 - controlled by concentration / uniform convergence
- Approximation error
 - due to restricted hypothesis class $\mathcal{F}$
 - decreases as $\mathcal{F}$ becomes richer

Expectation Identities

- Law of total expectation:

$$\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid Y]]$$

- In learning:

$$\mathbb{E}_{\mathbf{Z} \sim p^1}[\mathcal{R}(\hat{f})] = \mathbb{E}_{\mathbf{Z} \sim p^1}[\mathbb{E}_{(\mathbf{x}, y) \sim p}[\mathcal{L}(y, \hat{f}(\mathbf{x})) \mid \mathbf{Z}]]$$

- Law of total variance:

$$\text{Var}(X) = \mathbb{E}[\text{Var}(X \mid Y)] + \text{Var}(\mathbb{E}[X \mid Y])$$

- Interpretation:
 - randomness can be decomposed into conditional and marginal parts
 - key tool for analyzing learning algorithms

Concentration (Hoeffding Inequality)

- Let $X_1, \dots, X_l$ be i.i.d. with $X_i \in [a, b]$
- Hoeffding inequality:

$$\mathbb{P}\left(\left|\frac{1}{l} \sum_{i=1}^l X_i - \mathbb{E}[X_1]\right| \geq \varepsilon\right) \leq 2 \exp\left(-\frac{2l\varepsilon^2}{(b-a)^2}\right)$$

- Application to learning:

$$X_i = L(y_i, f(\mathbf{x}_i)) \in [0, 1]$$

- Then for fixed f :

$$\mathbb{P}(|R_{\text{emp}}(f) - R(f)| \geq \varepsilon) \leq 2e^{-2l\varepsilon^2}$$

Modes of Convergence

Let $(X_n)_{n \in \mathbb{N}}$ be a sequence of random variables and X a random variable

- **Convergence in probability:** $X_n \xrightarrow{P} X \iff \forall \varepsilon > 0 : \lim_{n \rightarrow \infty} \mathbb{P}(|X_n - X| > \varepsilon) = 0$
- **Almost sure convergence:** $X_n \xrightarrow{\text{a.s.}} X \iff \mathbb{P}(\lim_{n \rightarrow \infty} X_n = X) = 1$
- **Convergence in distribution:** $X_n \xrightarrow{d} X \iff \lim_{n \rightarrow \infty} F_{X_n}(t) = F_X(t)$ for all continuity points t of F_X , where $F_X(t) = \mathbb{P}(X \leq t)$
- **Relations:** $X_n \xrightarrow{\text{a.s.}} X \Rightarrow X_n \xrightarrow{P} X \Rightarrow X_n \xrightarrow{d} X$
- **In learning:** $R(\hat{f}_n) \xrightarrow{P} R(f^*)$

Necessity of Assumptions



Necessity of Assumptions

Theorem (No Free Lunch)

For every learning algorithm $A : \mathcal{Z}^l \rightarrow \mathcal{F} \subseteq \{f : X \rightarrow \{-1, 1\}\}$ there exists a distribution p on $X \times \{-1, 1\}$ with

$$\mathbb{E}_{\mathbf{Z} \sim p^l} [\mathcal{R}(A(\mathbf{Z}))] - \mathcal{R}(f^*) \geq \frac{1}{2}.$$

Without assumptions on the data distribution, learning is impossible.

Werner Zellinger

Institute for Machine Learning



☎ +43 732 2468 4532
✉ @zellinger@ai-lab.jku.at
🌐 jku.at/ai-lab-team